

REVERSEBIDS.NET

Real-Time B2B Reverse Auction Platform

PROJECT REPORT

Super-Lean Real-Time Reverse Auction Engine

MVP 1.0 — Architecture, Stack & Delivery Plan

Prepared for	Goodrich Ingredients and Chemicals Ltd., Chennai
Initiative	ReverseBids.net — Fishmeal & Fish Oil Procurement
Pilot scope	12 Buyers · 50 Suppliers · 2 SKUs
Document	MVP 1.0 Technology & Delivery Blueprint
Status	For internal review

Contents

1. Executive Summary

This report sets out a lean, low-risk technology blueprint for delivering ReverseBids.net MVP 1.0: a working, reliable, real-time reverse auction engine. It is presented as an alternative evaluation track — a deliberately low-code path explored alongside the heavier custom-engineered build — rather than a final commitment. At this stage the objective is not to build another full-scale e-commerce platform, but to prove a single thing convincingly: that real-time reverse auctions function reliably with real buyers and suppliers under live conditions.

The strategic premise is that market proving — not cost or engineering depth — is the single binding bottleneck for version 1.0. Until live demand is demonstrated, heavy investment is premature. Version 1.0 therefore optimises ruthlessly for time-to-market and cost efficiency; from version 1.1 onwards, with the market validated, cost ceases to be the governing constraint and capability can scale freely.

The recommended approach pairs FlutterFlow for the user interface with Google Firebase (Firestore, Cloud Functions, Authentication, Cloud Messaging, Storage, and Hosting) for the backend. Firestore's native real-time synchronisation removes the single hardest piece of conventional auction engineering — maintaining a live, consistent bid state across many concurrent participants — without operating WebSocket servers or managing infrastructure.

Scoped to the pilot of 12 buyers, 50 suppliers, and 2 SKUs, this stack is realistically deliverable in approximately two weeks by one capable AI-assisted full-stack developer with part-time UI and QA support, at an estimated 130–190 person-hours. Critically, the proprietary business logic — the auction algorithm, qualification rules, tie-breaking, anti-sniping, and winner selection — is implemented in Cloud Functions so that it remains portable and protected as intellectual property, independent of any no-code tooling.

Bottom line

- The bottleneck is market proving — not cost or engineering depth.
- v1.0 optimises for speed and cost; from v1.1, cost is no longer the constraint.
- Stack: FlutterFlow + Firebase, with proprietary logic isolated in Cloud Functions.
- Build engine: Claude Opus 4.8 and supporting AI apps — ~2 weeks, ~130–190 person-hours.

2. Evaluation Stance & Build Philosophy

2.1 Position of this report

This blueprint is one of two parallel lenses on the same goal, evaluated before further commitment — not a competing build. It deliberately explores a low-code, fully managed path (FlutterFlow + Firebase) as the fastest and most cost-effective route to a market-provable product. It sits alongside the heavier, custom-engineered track — the FastAPI backend with JWT authentication, WebSocket live leaderboards, Redis pub/sub, Alembic migrations, and a pytest suite — advanced under the engaged development vendor on a fixed-price, twelve-

week footing. The purpose here is to weigh the low-code alternative on its merits: how quickly and cheaply it can put a working reverse auction in front of real users.

2.2 The bottleneck is market proving

For version 1.0, cost and engineering depth are not the constraint. The single binding bottleneck is market proof — demonstrating that buyers and suppliers will genuinely transact through live, real-time reverse auctions. Everything else is secondary until that is established. Version 1.0 therefore accepts pragmatic low-code trade-offs in exchange for reaching real participants sooner, treating speed-to-evidence as the primary success metric.

2.3 Cost philosophy across versions

Version	Governing Priority	Posture on Cost
1.0	Speed to market & cost efficiency — prove demand	Tightly constrained; low-code, managed services only
1.1+	Capability, scale, robustness — build on validated demand	Constraint relaxed; cost is no longer the limiting factor

2.4 AI-augmented build model

The engine behind this delivery model is an AI toolchain led by Claude Opus 4.8, supported by a small set of complementary AI applications, used for code generation, debugging, documentation, and testing. This is what compresses a conventional multi-week effort into a roughly two-week sprint and underpins the low person-hour estimate: the developer orchestrates AI across each layer rather than hand-writing it all. In practical terms, the AI stack functions as the working back end of the build itself — the force multiplier that makes a lean, low-cost version 1.0 realistic.

Why this matters

A low-code front (FlutterFlow + Firebase) plus an AI-led build engine (Opus 4.8 and supporting apps) attacks the one thing that counts at this stage — getting to market proof fast and cheap. Depth and scale are a deliberate v1.1 concern, not a v1.0 one.

3. Strategic Objective & Scope

The guiding principle for MVP 1.0 is focus. The platform's value is not in breadth of features but in the trustworthy operation of its core mechanic. Every architectural decision below is made to reach a working, observable reverse auction as quickly and cheaply as possible, while keeping a clean path to scale.

3.1 What MVP 1.0 must prove

- Suppliers can submit competing bids that update in real time for all authorised participants.
- The engine applies qualification rules, ranks bids correctly, and selects a winner deterministically.

- Anti-sniping auto-extension behaves predictably as auctions approach close.
- Every action is captured in an auditable trail suitable for procurement governance.

3.2 Pilot envelope

Dimension	MVP 1.0 Target
Buyers	12 procurement participants
Suppliers	50 registered bidding suppliers
Product SKUs	2 locked SKUs (fishmeal / fish oil category)
Auction model	Lowest qualified bid wins, protein-normalised where applicable
Geography	Initial pilot — Tamil Nadu & Andhra Pradesh supply base

4. Recommended Lean AI Technology Stack

The table below maps each layer of the platform to a recommended technology and the rationale for the choice. The unifying theme is that Firebase provides a managed, real-time backend, while FlutterFlow accelerates front-end delivery across web and mobile from a single project.

Layer	Recommended	Why
UI	FlutterFlow	Very fast development; mobile and web from one project.
Database	Firebase Firestore	Native real-time updates; excellent fit for live auctions.
Authentication	Firebase Auth	Email, Google, and OTP sign-in with minimal coding.
Backend Logic	Firebase Cloud Functions	Bid validation, auction close, and winner calculation.
Notifications	Firebase Cloud Messaging	Outbid alerts, closing reminders, winner notifications.
File Upload	Firebase Storage	RFQs, certificates, and technical specifications.
Admin Dashboard	FlutterFlow Web	Manual moderation and auction control.
AI Assistant	Claude / ChatGPT / Gemini	Code generation, debugging, documentation, testing.
Source Control	GitHub	Versioning and collaboration.
Hosting	Firebase Hosting	Deploy in minutes.

5. Architectural Rationale

For ReverseBids.net at pilot scale, this architecture is close to ideal because it removes nearly all of the infrastructure burden that normally dominates the build and operation of an auction system. The managed Firebase platform eliminates entire categories of work:

▼ No server management	▼ No Linux administration
▼ No MySQL tuning	▼ No REST API boilerplate
▼ No WebSocket server	▼ No Kubernetes
▼ No DevOps complexity in the MVP	▼ Managed scaling out of the box

The decisive advantage

Firestore itself provides the real-time synchronisation needed for auctions. Bid updates propagate to every connected client automatically, removing the need to build, host, and scale a custom WebSocket layer — historically the riskiest part of a real-time auction system.

6. Core Intellectual Property — Custom Business Logic

No-code convenience must stop at the boundary of the business logic. The defensible value of ReverseBids.net lives in its proprietary rules, and these must be implemented in Cloud Functions — written, owned, and version-controlled by the business — so they remain portable across any future platform change.

Proprietary Component	Role in the Engine
✓ Reverse auction algorithm	Core mechanic governing how competing bids drive price down.
✓ Bid qualification rules	Determines which bids are valid and eligible to win.
✓ Tie-breaking logic	Resolves equal qualified bids fairly and deterministically.
✓ Auto-extension (anti-sniping)	Extends close time on late bids to ensure fair competition.
✓ Winner selection	Applies lowest-qualified-bid logic to declare the outcome.
✓ Procurement intelligence	Context and insight layered onto raw bid data.
✓ Audit trail	Immutable record of every action for governance and trust.
✓ Supplier ranking	Performance and reliability scoring across auctions.
✓ Analytics	Pilot metrics and reporting on auction behaviour.

Implementing these in Cloud Functions keeps the intellectual property independent of FlutterFlow and Firestore-specific tooling. Should the platform later migrate or scale onto different infrastructure, the logic moves with it.

7. Implementation Timeline — 14-Day Sprint

The following two-week schedule assumes one capable full-stack developer using AI assistance extensively, with part-time UI and QA support. It moves deliberately from interface to data model to the auction engine, leaving the final days for hardening and pilot deployment.

Days	Focus	Outcome
1-2	UI screens	Buyer, supplier, and admin views
3-5	Authentication + Firestore data model	Sign-in and live data foundation
6-8	Reverse auction engine + bid validation	Working core mechanic
9-10	Notifications + dashboards	Alerts and control surfaces
11-12	Testing and bug fixing	Stabilised build
13-14	Deployment and pilot	Live pilot launch

8. Effort & Resourcing Estimate

FlutterFlow, Firebase, and AI assistance together remove much of the infrastructure work that inflates a traditional build, bringing the total estimate substantially below a conventional custom project.

Role	Effort
Full Stack Developer	80-120 hours
UI/UX Designer	20-30 hours
QA Tester	20-30 hours
DevOps	8-16 hours
Total	~130-190 person-hours

9. Deferred Scope — Version 2.0

To protect the two-week timeline and the clarity of the core proof, the following capabilities are deliberately postponed until after the pilot validates the engine. None are required to demonstrate that real-time reverse auctions work reliably.

▼ Payment gateway	▼ ERP integration
-------------------	-------------------

▼ Advanced analytics	▼ AI price prediction
▼ Multi-language support	▼ Complex approval workflows
▼ Multiple product categories	▼ Buyer-side budgeting tools

10. Architectural Enhancement — Path to Scale

One enhancement is worth designing in from day one, even though it is not built during the MVP. Structuring the data flow so that it can later feed an analytics warehouse gives a clean path from a lean pilot to enterprise-scale intelligence without redesigning the platform.



As auctions grow from 12 buyers and 50 suppliers to hundreds or thousands, capabilities can be added progressively while the core architecture stays intact. BigQuery remains optional and dormant until analytics volume justifies it — but designing the pipeline to accommodate it now avoids costly rework later.

11. Conclusion & Recommendation

For the current goal — launching a working real-time reverse auction MVP in about two weeks — the FlutterFlow plus Firebase stack is one of the fastest, lowest-risk, and most cost-effective options available today. It eliminates infrastructure overhead, provides real-time synchronisation natively, and concentrates effort where it matters: proving the engine.

The single most important discipline is to keep the proprietary auction logic in portable Cloud Functions, so the intellectual property is never trapped inside no-code tooling. With that boundary respected, ReverseBids.net can validate its core mechanic quickly and cheaply, then scale on a clean, pre-planned path toward enterprise analytics.

Recommendation

Adopt this low-code track as the route to version 1.0, with the explicit aim of proving the market fast and cheaply using FlutterFlow + Firebase and an AI-led build engine (Opus 4.8 and supporting apps). Keep all proprietary logic in portable Cloud Functions, and design the Firestore-to-BigQuery pipeline now. Hold deeper engineering and cost for version 1.1, once live demand is validated — because at this stage, market proof is the only bottleneck that matters.